Analysis of extreme flooding events through a calibrated 1D/2D coupled model: the case of Barcelona (Spain)

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ABSTRACT

This paper presents the results of a calibrated 1D/2D coupled model simulating surface and sewer flows in Barcelona. The model covers 44 km^2 of the city land involving 241 km of sewers. It was developed in order to assess the flood hazard in the Raval district, historically affected by flooding during heavy rainfalls. Special attention was paid to the hydraulic characterization of the inlet systems (representing the interface between surface and underground flows), through experimental expressions used to estimate the runoff amounts effectively introduced into the sewers in case of storms. A 2D unstructured mesh with more than 400,000 cells was created on the basis of a detailed DTM. The model was calibrated and validated using four sets of well-recorded flooding events occurred in 2011. The aim of this paper is to show a detailed 1D/2D coupled model can be adequately calibrated and validated using a wide set of sewer sensors and post events collected data (videos, photos, emergency reports, etc.). Moreover the created model presents significant computational time savings via parallel processing and hardware configuration. Considering the computational performances achieved, the model can be used for real time strategies and as the core of early warning systems.

KEYWORDS

1D/2D coupled modelling, dual drainage, sewer and overland flows calibration.

INTRODUCTION

The EU 7th Framework Programme CORFU Project (Collaborative Research on Flood Resilience in Urban Areas, www.corfu7.eu) is an interdisciplinary project funded by the European Commission with the aim to improve the practice of urban flood risk management. In this context, eight case studies in Europe and Asia have been developed with the main objective to analyze and improve resilience of cities to flood impacts. One of the case studies is Barcelona in Spain. The average annual precipitation in Barcelona is 600 mm, but the 5-minute 10-year design storm intensity is 204.7 mm/h and it is not rare that 50% of the annual precipitation occurs during two or three rainfall events. The morphology of Barcelona presents areas close to Collserola Mountain with high gradients (with an average of 4% and maximum values of 15-20%) and other flat areas near to the Mediterranean Sea with mild slopes (less than 1 %) and several spots susceptible to flooding. This morphology produces flash floods in the bottom part of the city in case of heavy storm events. Due to the adverse rainfall patterns, the morphological characteristics and the high density of population and impervious surface, it is clear that drainage in Barcelona has special relevance.

In the nineties, a specific company responsible of the management of the Barcelona sewer system, Clavegueram de Barcelona S.A. (following CLABSA), was created and the drainage of the city was transformed in order to be more effective against floods and water pollution. With two main tools, – hydraulic regulation and real time control – Barcelona is nowadays one of the cities with the most advanced management of drainage systems. However, notwithstanding the big efforts carried out in the last two decades (several sewer pipes and facilities were re-designed and improved for a 10 years return period), flooding problems still occur in case of heavy storm events causing damages to assets and people in some critical points of the city. These flooding problems are produced by sewer floods and by the runoff that is not collected by the sewers due to a poor surface drainage capacity.

A specific area of Barcelona, the Raval district, was selected as case study, considering, for the whole area, a flood risk assessment based on the hazard and vulnerability evaluation. The Raval district, with almost 50,000 inhabitants in an area of 1.09 km^2 is one of the most densely populated areas in Europe

(approx. 44,000 inh./km^2). The district area is highly impervious with several vulnerable elements (such as schools, hospitals, major highways etc.). It represents a critical point of the city where stormwater not conveyed into the sewer network and overflows from sewer manholes are stored (Figure 1). Moreover, the hydrological response time of Raval district catchment is very short (less than 30 minutes).

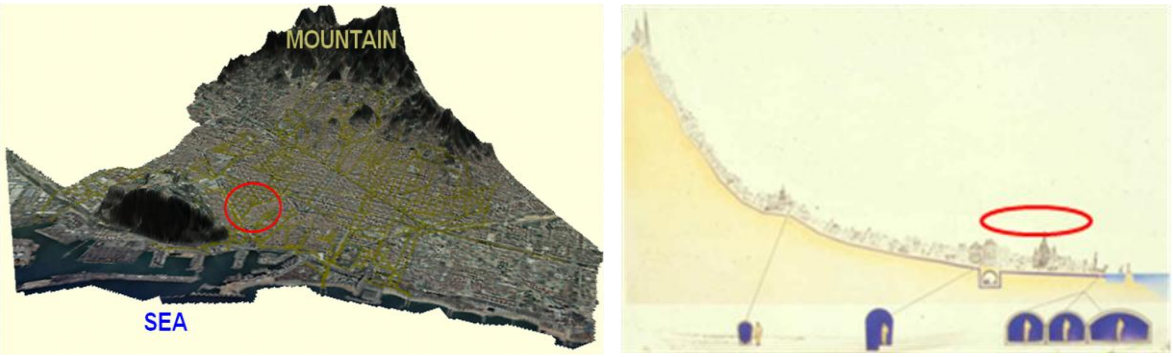


Figure 1. Typical gradients and morphology in Barcelona. The circles in both images indicate the location of Raval District.

Literature review suggests a very wide range of methods depending on how detailed the analysis needs to be in order to give a realistic representation of the hydraulic behaviour of the analysed catchment. The methods used in research and practical applications range from very basic approaches with numerous simplifying assumptions up to very sophisticated ones with high resolution data and calculation time demanding (Apel *et al*., 2009; Aronica and Lanza, 2005).

Traditional 1D sewer models used by CLABSA during recent years were unable to adequately describe flooding in the Raval district and the causes producing these problems (great runoff volumes not conveyed into the sewers coming from upstream catchments). Even if it is clear that the choice between using a 1D surface network model or a 2D surface system model determines the accuracy of results and the computational time required to obtain them, when the flow overtops the curbs in the streets and it does not remain within the street profile, using a 2D model is crucial (Mark *et al*. 2004). Considering that Raval district is located in a very flat area and that many urban areas are occupied by pedestrian areas and squares (without curbs and preferred runoff direction), it is evident the need of a

detailed coupled 1D/2D approach in order to correctly take into account surface flows coming from upstream catchments and the interactions between the two drainage layers (known, respectively, as 'major system' formed by streets, sidewalks, squares, etc. and 'minor system' formed by the sewer network). Moreover, the high information availability for the Barcelona case study in terms of sewer elements (such as pipe sections, topographic data of manholes and inlets, hydraulic performance of inlets systems, etc.), rainfall series data provided by a dense rain gauge network, water depth series provided by distributed sewers sensors, as well as the new advances in parallel processing and hardware (Brodtkorb *et al*. 2012; Henonin *et al*., 2013; Lamb *et al.*, 2011; Kalyanapu *et al*., 2011; Satra and Brodtkorb, 2012), have encouraged the setup and calibration of a detailed 1D/2D coupled model.

This paper shows the main results and features concerning the creation and the calibration of a detailed 1D/2D coupled model able to provide realistic simulations of flooding produced by heavy storm events in the Raval district and the elaboration of hazard maps on the basis of the local estimation of the flow parameters (flow depth and flow velocity). Particularly, the model was calibrated and validated using a wide set of rainfall time series, water depth time series coming from a distributed sensors network and post events collected data (videos, photos, emergency reports, etc.). Moreover speeding up strategies focused on parallel processing and hardware advances were implemented in the model reaching very high computational performances that allow its use for real time strategies and as the core of early warning systems.

METHODS

General aspects for the creation of a coupled 1D/2D urban model

Recently several authors published interesting papers treating the need of coupled approaches (modelling of the surface and sewer flows at the same time) to represent adequately urban flooding caused by sewer overflows (Phillips *et al*., 2005; Lipeme Kouuyi *et al*., 2008; Obermayer *et al*., 2010;

Leandro *et al*., 2009) and carry out a realistic flood risk assessment (Kandori and Willems, 2008). Currently, a 1D/2D coupled model represents a powerful tool to describe, in a very realistic way, the hydraulic behaviour of urban areas suffering flooding problems due to the excess of runoff not conveyed by the drainage networks. However, in order to ensure good and detailed results, it is not sufficient to have a technologically advanced tool for the calculation concerning hydrological and hydraulic processes, but several other aspects must also be carefully considered, such as:

- Detailed topographic data able to reproduce complex urban morphologies
- Methodologies able to take into account flow interchanges between the two layers of drainage (surface and underground layers)
- Rainfall and flow / flow-depths records for the adjustment of the model parameters in the calibration phase and for the results verification
- Time series of the operations carried out by the sewer devices (variable sluices, pumps, etc.)

These aspects were considered for the creation of the 1D/2D coupled model of the Raval District. The software used for the creation of the model was the version 3.5 of Infoworks ICM by Innovyze (2012).

Topographic data

Urban areas present a very complex topography due to the presence of elements such as buildings, sidewalks, roadways, gutters, walls and banks. When a storm event occurs in our cities, generally, runoff produced in roofs and terraces is directly conveyed to the underground sewer networks, while runoff produced in roadways, parks, squares, etc. circulates over the urban surfaces until it reaches the inlet structures of the drainage systems. Moreover, due to relatively low roughness of the surfaces in urban areas, stormwater runoff is characterized by low flow depths (often less than 15-20 centimetres) and high flow velocities (up to 3-4 meters per second). In this context, it is crucial to have a detailed Digital Terrain Model (DTM) with high resolution. For this study, a specific DTM model was provided by the Catalan Institute of Cartography. This DTM presents a resolution of 1 m^2 obtained by

a LIDAR (*Laser Imaging Detection and Ranging*) flight with a density of 1.33 points/m^2 and a precision of 15 cm in terms of ground elevation. This DTM was also improved using data concerning CLABSA territorial system (for example positions and elevations of manholes, inlets, etc.).

Characterization of the hydraulics of surface drainage systems

The need to estimate the hydraulics of the surface drainage structures and to model them through a coupled 1D/2D approach is clear. Moreover, the possibility to characterize hydraulically all the connections (manholes and drain inlets) between surface and underground systems allows to improve the estimation of surface and pipe flows during a storm. The consideration, at the same time, of the surface flow and its interaction with sewer flow is commonly known as '*dual drainage modelling*', with flow components on the surface and underground (Djordjevic *et al*., 1999) (Figure 2).

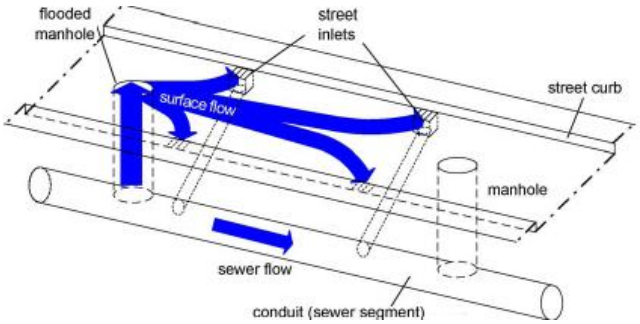


Figure 2. Interaction of surface and sewer flow (dual drainage concept) (Schmitt et al. 2004).

In order to achieve this objective, the hydraulic performance of the inlets present in the Raval District was analyzed using experimental expressions proposed and implemented by Gómez and Russo (2009; 2011) and by Russo *et al.*, (2013). Specifically, all the inlets located in the Raval District (more than 2600 units of almost 50 different types) were classified in groups represented by 8 inlet type previously studied by Gómez and Russo (2011) (Figure 3). The hydraulic efficiency of these grated inlets was analyzed according to the specific type and several geometric parameters of the location (longitudinal and transverse slopes, gutter section, width of the roadway, etc.). Finally, hydraulic

efficiency of each grated inlet was compared to the performance of the Barcelona1 inlet (the top right one in Figure 3) in order to obtain specific weighted efficiency coefficients defined as following:

$$c_{we_i} = \frac{E_i}{E_{Barcelona1}} \quad (1)$$

where:

c_{wei} is the specific weighted efficiency coefficient for the type grate i (from 1 to 8).

$E_{Barcelona1}$ is the hydraulic efficiency of the grated inlet Barcelona1 (where efficiency is defined as the ratio between the flow captured by the grated inlet and the flow circulating in the gutter).

E_i is the hydraulic efficiency of the grated inlet type i.

Combining street geometric configurations and the types of grates c_{wei} range varies from 1 to 2.1.



Figure 3. Grated inlet types considered for the flood hazard assessment study in the Raval District. The Barcelona 1 inlet is shown in the top right image (framed with a continuous red line).

Once the hydraulic performance of each inlet was estimated through empirical expressions, a further efficiency reduction was applied using a clogging factor of 50% that was uniformly applied to all inlet types and all geometric configurations in order to take into account possible reduction of the hydraulic efficiency due to blockage of grated inlet void areas during storm events (Gómez and Russo, 2013).

Creation of the 1D/2D coupled model

Dual drainage modelling cannot be adequately represented without a careful consideration of hydraulic efficiency of surface drainage structures (inlets, transversal grates, etc.) and an adequate representation of the urban topography (buildings, streets and sidewalks, curbs, etc.). Therefore a powerful informatics tool is necessary to simulate all the hydrological and hydraulic phenomena involved during heavy storm events. As mentioned, a two dimensional simulation is better suited than a one-dimensional simulation for modelling flows through complex geometries (such as urban streets and buildings, road intersections and other transport infrastructure). The same happens in open ground, where either source or direction of flow is not pre-defined. In urban areas the situation is exacerbated further by the presence of sewer networks, in which flows can both enter and exit the system during flood events. It is clear that modelling such complex systems requires coupled 1D and 2D engines to provide accurate results.

The software Infoworks ICM by Innovyze (following ICM) (2012) was adopted to assess flood hazard in the Raval District during storm events. ICM solves the complete 2D Saint Venant equations in a finite volume semi-implicit scheme (Godunov, 1959) with a Riemann solver (Alcrudo and Mulet-Martí, 2005). ICM combines a number of distinctive features as the analysis and prediction of potential flood extent, depth and velocity, and the modelling of the interaction of surface and underground systems in a fully integrated environment. Moreover, it allows the creation of multiple surface unstructured meshes that optimize modelling flexibility and accuracy.

In order to consider surface and sewer flows coming into the Raval District from upstream catchments, an extended area was considered in the study. Only main sewers were considered for these catchments, while main and secondary networks were taken into account for Raval District. The final model considered a total area of 44 km^2 with 3874 nodes, 241 km of total pipe length and 6 major storage facilities with a total capacity of 170,000 m^3 (Figure 4).

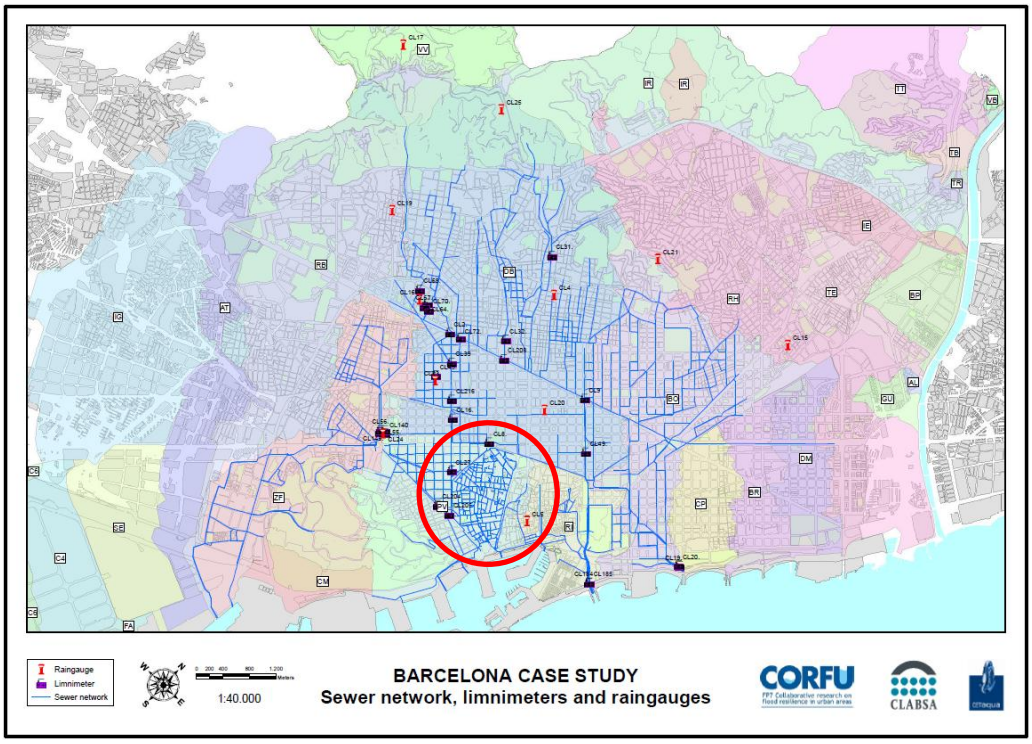


Figure 4. The final model of Barcelona case study. The circle highlights the high density of pipes and nodes corresponding to the main and secondary networks of the Raval District.

A 2D mesh covered the whole analyzed domain with 403,822 triangles. The mesh was created using Shewchuk meshing algorithm (Shewchuk, 1996). The algorithm enables creation of an irregular mesh based on the main parameters: average mesh size, the maximum cell size, minimum angle. In this case these parameters were fixed to 25 m^2, 100 m^2 and 25º respectively, even if for narrow streets and complex topographies (e. g. in the Raval) the cell size decreases significantly to adapt the mesh to the 2D domain. The ground level for a mesh element is calculated by sampling the ground model within the 2D triangles making up the element and then taking the average of the sample point levels.

The number of sample points for each triangle is determined by subdividing the triangle until the minimum element area or, when using a Gridded Ground Model, the ground model resolution is reached. The sample points are the centroids of the resulting triangles. So, for the proposed model the terrain data were integrated into the model through the mesh cells, even if 48 break lines / walls were defined and introduced during the 2D meshing to reproduce urban elements as walls, banks, etc.

Parks and other green areas were represented in the same 2D mesh, trough "2D infiltration zones" characterized by their specific hydrological, physical and geometric parameters (Figure 5), while buildings (formed by roofs ad courtyards) were represented as void areas (Schubert and Sanders, 2012). Spatial rainfall distribution was applied to 2D domain and subcatchments containing roofs and courtyards according to Thiessen polygons method. For each 2D element and each subcacthemnt, the code ICM checks in which rain gauge region (if any) the centroid of the element is located and applies the rainfall profile accordingly. Runoff produced in the building areas was estimated considering an approximation of single non-linear reservoir (whose routing coefficient depends on surface roughness, surface area, ground slope and catchment width) and directly conveyed into the sewer network. This goes in accordance with local practice in Barcelona, where roofs, terraces and courtyards (approximately corresponding to 50% of the whole analyzed domain) are directly connected to the underground sewers.



Figure 5. Flexible unstructured mesh with triangular cells representing the streets and pervious areas (shown with shadows) of the city. Building areas (with roofs and courtyards) are excluded by 2D domain. On the bottom-right corner of the figure, it is possible to see the narrow streets of the Raval District involved in the 2D mesh.

ICM allows the hydraulic characterization of inlets and the consequent possibility to carry out 1D/2D simulations according to the dual drainage concept previously explained. The inlet effect in the model is to limit the flow rate from the surface into the manhole, or vice versa. This can lead to the situation (as occurs in reality) where there is surface flooding whilst there is available capacity in the below ground system.

Nodes are used in 2D simulations to model the exchange of flow between the sewer system and a 2D meshed area (Figure 5). Surcharging of a manhole results in an inflow to the 2D mesh element in which it resides. Calculation of flow exchange between the manhole and the mesh depends on which "*flood type"* is chosen:

- *2D*: exchange of water between the 2D manhole and the mesh is calculated using weir equations assuming a weir crest level at the node ground level and crest length equal to the node shaft circumference. A flooding discharge coefficient equivalent to the weir discharge coefficient is specified for the node.
- *Gully 2D*: Exchange of water between the 2D manhole and the mesh is calculated using a specific "*Head/Discharge relationship*".

In the proposed model, the manholes present either flood type "2*D"* or "*Gully2D"*. Specifically, the manholes outside of the Raval District were characterized as with a "*2D*" flood type. In this area only the main network was considered for the model and, consequently, inlet system capacity was neglected (flow interchanges occur through manholes regardless of inlet system capacity). In this case a flooding discharge coefficient of 0.5 was adopted according to several papers that suggest this value to characterize the interaction between surface and sewer flows (Leandro *et al*., 2007 and 2009; Djordjević *et al*., 2013). The manholes inside the Raval District were characterized through a "Gully 2D" flood type. In this case several curves *Head vs. Discharge* (Figure 7) were elaborated in order to take into account the hydraulics of the inlets located in the Raval District according to the cited methodology (Gómez and Russo, 2011). These curves represent two possible situations: the first one considering flow exchange from 2D surface to 1D sewer due to a "positive" head gradient (i.e. when

water level on the surface is above the water level in the manhole, Figure 6 left), and the other one representing flow exchange between 1D sewer to 2D surface due to a "negative" head gradient (i.e. when piezometric head in the manhole is above the water level on the surface due to high pressure in the manhole, Figure 6 right). In the first case the experimentally derived expressions were used to estimate intercepted flow while in the second case a discharge coefficient of 0.5 was used according to the previous considerations.

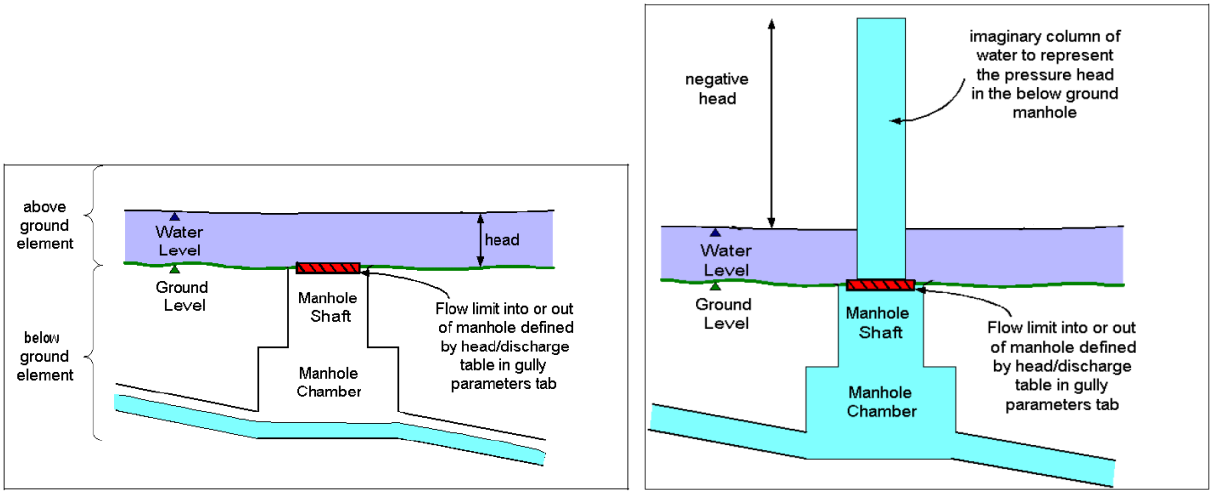


Figure 6. Modeling of flow interchanges between surface and the sewer system are represented. On the left, flow arrives at the manhole from the subcatchment or the above surface flow whilst there still is capacity available in the below ground sewer (but there is a limit to how much flow can enter the sewer imposed by the inlet). On the right, water level in the below ground sewer rises and flows out on to the catchment surface (but there is a limit to how quickly it can do this, again imposed by the inlet).

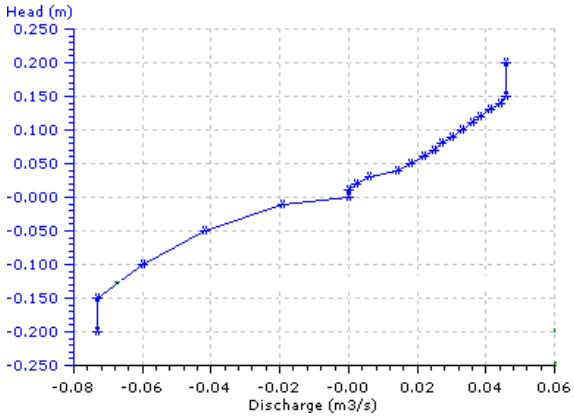


Figure 7. Head/Discharge curve for a node with "Gully 2D" flood type to model the flow exchanges between the sewer system and the 2D meshed area.

As mentioned, flow interchanges were simulated through manholes (nodes) although they occur through the inlets too. ICM allows to treat a manhole as a set of inlets defining the "number of equivalent inlets" n_{eq} for each Gully2D. n_{eq} is a scaling factor used to multiply the discharge values of the *Head/Discharge Tables* and for this model was calculated using the following equation:

$$n_{eq} = \sum_{i=1}^{8} c_{wei} \cdot n_i \qquad (2)$$

where:

n_{eq} is the integer number of equivalent inlet associated to a Gully2D node

n_i is the number of inlet with type i (i=1, …, 8) associated to a Gully2D node.

Head/Discharge Tables for the inlet Barcelona1 were defined for several geometric conditions and were related to each Gully2D node.

In order to determine the number of inlets associated to each Gully2D node, the "microcatchments" were defined as the portion of the 2D surface (containing grated inlets) related to each Gully2D node. (Figure 8). These microcatchments were defined generating overland flow paths through GIS tools according to the following steps:

- Generation of a Digital Surface Model considering the height of the building in order to achieve the effective flow paths;
- Definition of “potential microcatchments” using ArcGIS Spatial Analyst and Watershed Delineation (ArcGIS v.10)
- Definition of “final microcatchments” through a specific tool created by CLABSA to associate each “microcatchment” to the manhole with the lowest level inside it. In case no manhole is found inside the mircocatchment, its area and limit are merged to the downstream microcatchment.

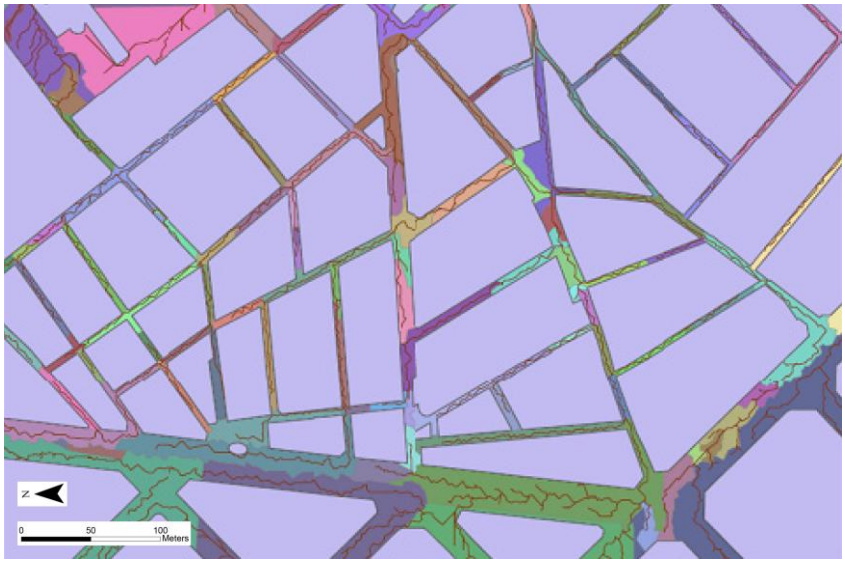


Figure 8. Overland flow paths and microcatchments in the Raval District.

RESULTS AND DISCUSSION

Calibration of the model

CLABSA Exploitation and Maintenance Department provided rainfall data for the calibration of the 1D/2D Raval model. Spatial rainfall distribution was applied considering rainfall data coming from 11 rain gauges of CLABSA network (Figure 4). Due to the recent significant changes of the Barcelona sewer system (new infrastructure such as pipes and tanks were built in the period 2008-2010), only

rainfall events in year 2011 were selected. According to literature, three storm events were selected for calibration process, while another one was selected for verification. The main characteristics of these events are shown in the Table 1.

Table 1. Events selected for calibration and verification of the model. The maximum values and the return periods T in parenthesis are referred to one of the 11 rain gauges of CLABSA network.

Date event	Cumulative rainfall (mm)	Maximum rainfall intensity in 20 minutes (mm/h)	Maximum rainfall intensity in 5 minutes (mm/h)	Event used for
15/03/2011	54.1	69.6 (T=1)	98.4 (T=0.5)	Calibration
07/06/2011	26.8	24.3 (T=0.2)	49.2 (T=0.2)	Calibration
19/07/2011	45.9	95.1 (T=5)	135.6 (T=2)	Calibration
30/07/2011	30.4	105.9 (T=8)	140.4 (T=2)	Verification

Flooding reports and 29 water level gauges located in manholes and conduits of the analyzed domain were used for the calibration/verification processes (Figure 1). Specifically, flow depth series calculated by the model were compared to flow depth series recorded by the water level gauges. Calibration results for a specific manhole located in Parallel Street are shown in Figure 9. The validation results for several points of the sewer network are shown in Figure 10.

Calibration and validation results show a good agreement for the data concerning heavy storm events (19-07-2011 and 30-07-2011), while for less extreme events (15-03-2011 and 07-06-2011) the differences increase, even if, in all the cases, the temporal evolution of the series is very similar and allow an adequate matching of the intermediate peaks. For example, for the node (P-AV65), the errors calculated as the difference between the measured and simulated flow levels respect to sewer invert level (7.95 m) are around 50% (for the less intense storm events of 15/03/2011 and 07/06/2011), while for the heavy storm events of 19/07/2011 and 30-07-2011 these difference are less than 5%.

In addition, flooding parameters provided by the model (perimeter of flooded areas, flow depths of the overland flow, etc.) were compared to observed data collected in the post-events emergency reports of policemen and firemen, amateur videos and photos recorded during the selected storm events (Figure 11 and Figure 12). In this way, it is possible to compare overland flow depth to reference elevation of objects located in surface (walls, cars, urban furnitures, etc).

The locations of sewers and surface points used for calibration and validation of the model are shown in the Figure 13. Moreover, it is possible to observe the good agreement between the historically flooded areas of the Raval and the results, in terms of flow depths map, of the model for a synthetic design storm with a return period of 10 years and a duration of 1 hour (Figure 14).

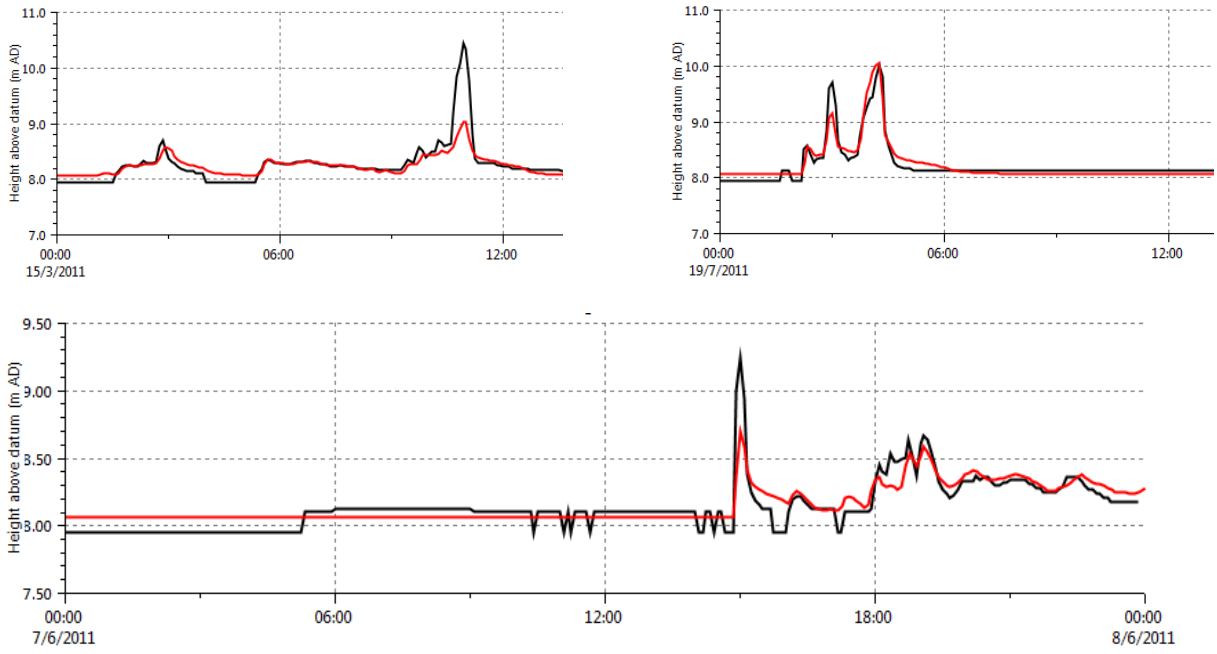


Figure 9. Calibration results (for the three selected events) regarding a monitored manhole (P-AV65) located in Parallel Street (Raval District). Black lines show recorded flow depths, while red ones show simulated flow depths. The errors, calculated as the difference between the measured and simulated flow levels respect to sewer invert level (7.95 m) are 56.8% (for the event of 15/03/2011, top left of the figure), 42.1% (for the event of 07/06/2011, top right of the figure) and 2.2% (for the event of 19/07/2011, down part of the figure).

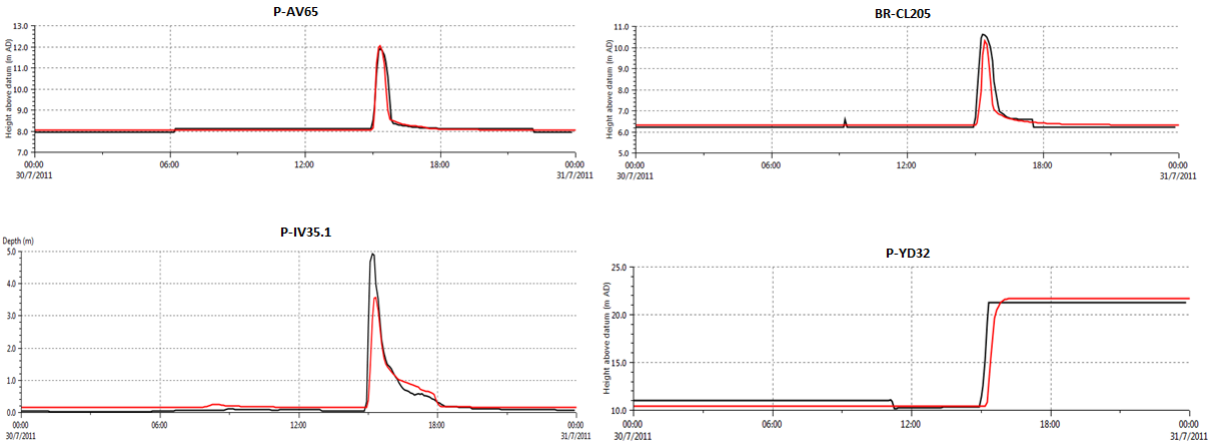


Figure 10. Validation results (event of 30-07-2011) for two manholes (P-AV65 and BR-CL205) in the Raval District (upper part of the figure), a pipe in Diagonal Avenue (P-IV35.1) and Urgell storage tank (P-YD32) upstream the Raval District (in the bottom part of the figure). Black and green lines show recorded flow depths, while red ones show simulated flow depths. The errors in the manholes and tanks, calculated as the difference between the measured and simulated flow levels respect to invert levels in terms of peak water level are 4.2% (for the P-AV65 manhole with invert level of 7.95 m), 7.2% (for the manhole BR-CL205 manhole with invert level of 6.22 m) and 1.7% (for the P-IV35.1 Urgell storage tank with invert level of 9.52 m). Error in pipe P-IV35.1, calculated as the difference in terms of flow depth, is 27.4%.

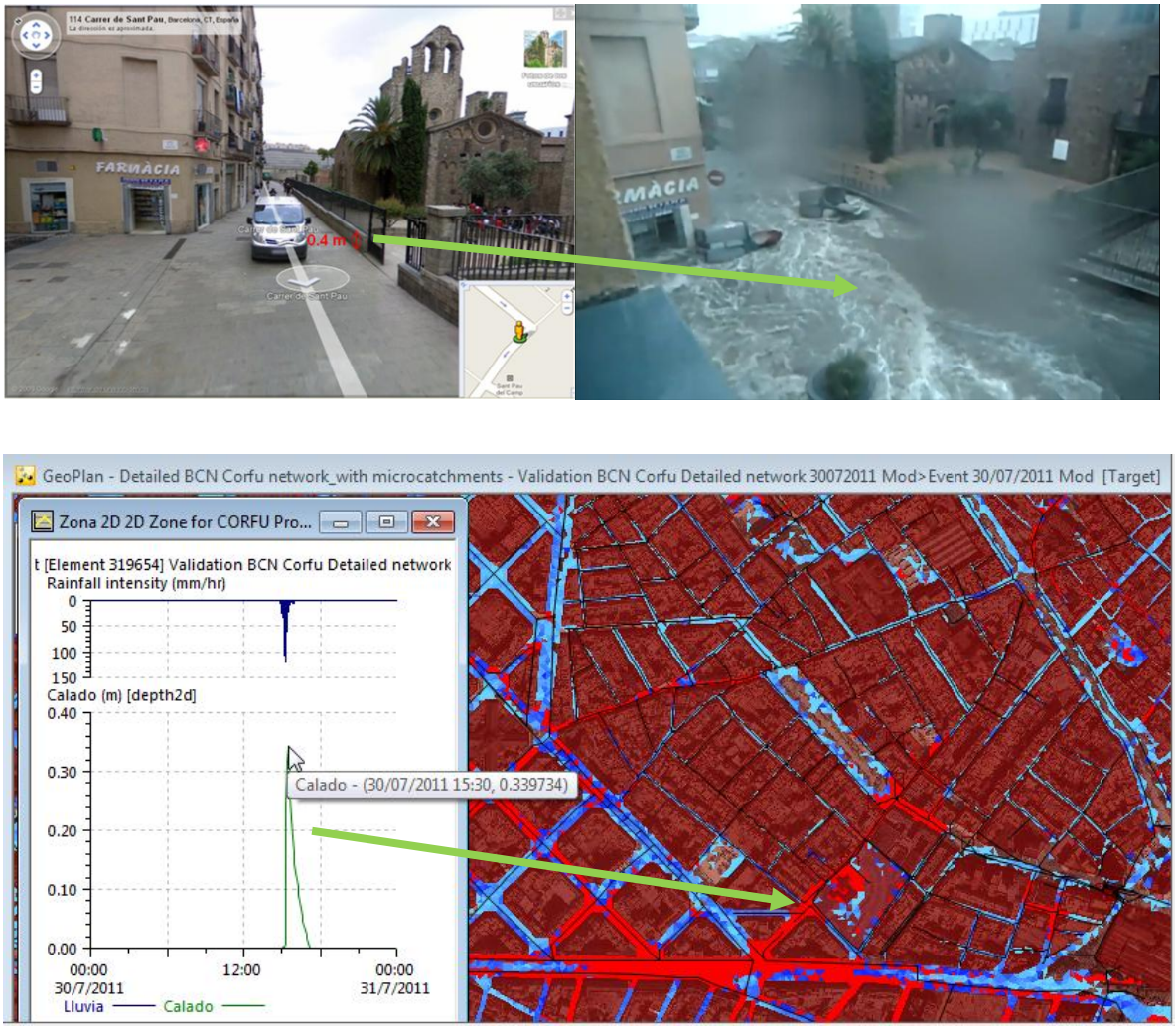


Figure 11. Validation results for Sant Pau Street (Raval District): flow depths provided by the model for the event of 30 July 2011 were compared to images and videos recorded during this event. Specifically, in the bottom of the figure, it is possible to observe the calculated peak flow depth (34 cm) in a cell near the Pharmacy located in the Sant Pau Street. In the upper part of the figure, on the left, a wall 40 cm high separates the street from the Sant Pau Church, while, on the right, an image of an amateur video shows the overland flow during the event.



Figure 12. Validation results for the crossroad between Diagonal Avenue and Casanova Street: Comparison between the profile representing surcharged pipes provided by ICM software and a photo taken during the event of 30 July 2011. Specifically, the maximum piezometric level corresponding to the photographed surcharged manhole was 46.6 cm.

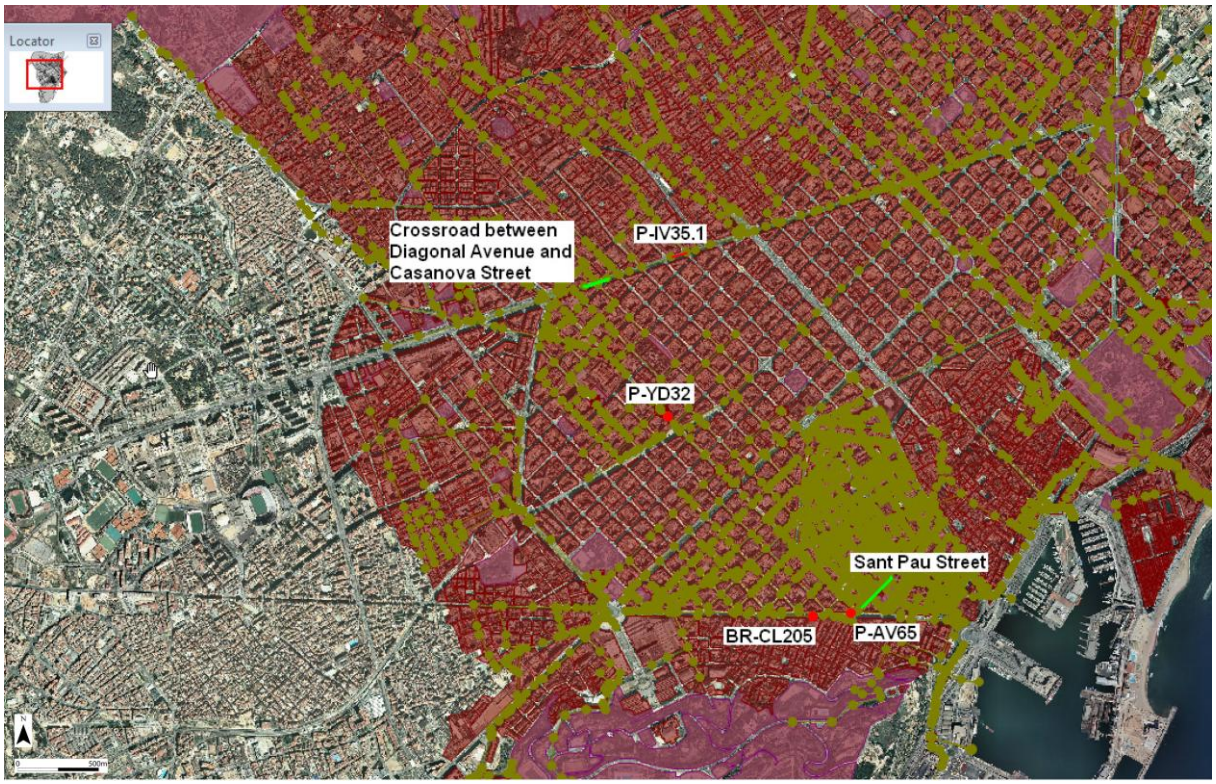


Figure 13. Locations of the sewers facilities and surface points used for the calibration and the validation of the model.



Figure 14. On the left, historical flooded areas inside the analyzed domain are remarked. On the right, hazard map in terms of maximum flow depths (represented through shadow areas progressively more intense) achieved through the 1D/2D coupled model for a storm with a return period of 10 years and historical flooded areas are shown.

Sensitivity analysis

1D/2D coupled models in urban areas represent an innovative technique to analyze the hydraulic behaviour of critical streets during heavy storm events. Available literature does not yet provide extensive reference on the importance and influence of all parameters involved in hydrological and hydraulic processes.

According to the results of a study developed by Sheffield University (Shephard *et al.* 2011) concerning the routing parameters in a 2D domain using Infoworks CS, the extent of the flooded area is relatively insensitive to mesh density and surface roughness, while the modelling of buildings has however been shown to have a significant effect on flood depths.

For Barcelona case study, sensitivity analysis was carried out in order to study the influence of Manning roughness (range from 0.013 to 0.020 $s/m^{1/3}$) on the surface flow. Results do not show strong dependency of the surface flood extension to this parameter.

The sensitivity of the model towards discharge coefficient was analyzed too. Literature on this parameter suggests to use values ranging from 0.5 up to 0.7 (Leandro *et al*., 2007 and 2009; Djordjević, 2013). Two different simulations with these extreme discharge coefficients showed that at global scale no change is detected for perimeters of the flooded areas, while at microscale (considering cells where flow interchanges occur) maximum differences in terms of flow depth ranged from 4 to 9%. It is useful to remark that discharge coefficients variation only affects flow interchange through the surcharged "*2D*" nodes located outside the Raval district. As previously explained, for the nodes located in the Raval, "Gully 2D" node type was selected and their hydraulic performance was estimated through checked experimental expressions (Gómez and Russo, 2009; Russo *et al*., 2013).

Computational time saving and configurations analysis

Historically, 1D/2D coupled models were considered too computationally demanding, so their use was restricted to those parts of the catchment where overland flow pathways could be multi-directional or could not be easily defined.

On the other hand, the required high-resolution topographic data for detailed simulations in urban areas is increasingly available from airborne altimetric LiDAR. Its wider application in large areas is limited by computational power and the performance of hydraulic modelling software. Recent researches focused on the following different techniques to reduce computation time of software models: dynamically adaptive grids, spatially-varying timesteps, simplified approximations for the physic phenomena considering kinematic and diffusive wave approximations (Liang *et al*., 2008; Sanders, 2008). The first options showed a relative computational time saving but introduced new criticisms and challenges. The latter, even if may allow faster computations, neglect significant aspects of flow.

In the past decade, modelling speeding-up strategies focus on parallel processing and hardware advances (Pau and Sanders, 2006; Sanders *et al.,* 2010). More recently, specific codes based on fast graphic processing units (GPU) usually dedicated to games and advanced computer graphics have been developed and used in fluid mechanics (Henonin *et al*., 2013; Lamb *et al*, 2011). Combining these features, it is possible to create hydrodynamic models capable of fully exploiting GPUs and CPUs with very promising performance in terms of computational time (Brodtkorb *et al*. 2012; Kalyanapu *et al*., 2011; Satra and Brodtkorb, 2012).

All these new advances were implemented in the Barcelona coupled model and a computational analysis was carried out considering the following hardware configurations for a specific simulation of 3 hours, with storm duration of 1 hour:

1. Workstation with 64-bit operating system (Windows 7) and 2Gb of RAM memory.
2. Virtual server in a blade system with 64-bit operating system (Windows 7), 2Gb of RAM memory and 4 CPU Intelxeon in order to exploit the ability of ICM to support full multi core processing.
3. Workstation with 64-bit operating system (Windows 7), 2Gb of RAM memory, 4 CPU Intelxeon for multi core processing and a specific GPU card that play an important role during the treatment of the 2D calculations.

The results of this analysis showed that run time vary from several days (6-7 days) for "Configuration 1", to several hours (5-6 hours) for "Configuration 2" and few minutes (3-4) minutes for the "Configuration 3".

It is clear that, since simulations can be carried out in few minutes, real time strategies and early warning systems could be activated on the basis of the results of this coupled model continuously running.

CONCLUSIONS

A detailed 1D/2D coupled model was developed to study the hydraulic behavior of a critical area of Barcelona (Raval district). The interface between the two drainage layers was characterized through empirical expressions related to hydraulic performance of surface drainage systems. The 2D domain covers 44 km^2 of the city land involving 241 km of sewers, while 2D mesh counts more than 400,000 triangular cells.

Calibration and validation of the model were based on a wide range of field data (rainfall and flow depth time series, amateur videos and photos, and post events reports provided by firemen and police) related to 4 heavy storm events that occurred in 2011. The obtained results show that it is possible to reproduce the effects of urban floods in the Raval district with a high level of detail.

The calibrated model can be used to assess flood hazard and risk and to estimate the economical losses produced during storm events. In this way, this could be a useful tool to design new infrastructures with clear objectives in terms of flood risk and economical loss reduction.

Moreover speeding up strategies focused on parallel processing and hardware advances were implemented in the model reaching very high computational performances that allow the use of this model for real time strategies and to be used as the core of early warning systems. Specifically, the model is capable of fully exploiting GPUs and CPUs with very promising performance in terms of computational time saving with run time of 3-4 minutes for complete simulations.

ACKNOWLEDGEMENTS

Research on the CORFU (Collaborative research on flood resilience in urban areas) project was funded by the European Commission through Framework Programme 7, Grant Number 244047.

Research on the PREPARED (PREPARED Enabling changes) project was funded by the European Commission through Framework Programme 7, Grant Number 244232.

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